- The acceleration due to gravity is denoted by $g\,\text{m s}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

13 A projectile is fired from ground level at $35\,\text{m s}^{-1}$ at an angle of $\theta°$ above the horizontal.

(a) State a modelling assumption that is used in the standard projectile model. **[1]**

(b) Find the cartesian equation of the trajectory of the projectile. **[4]**

The projectile travels above horizontal ground towards a wall that is 110 m away from the point of projection and 5 m high. The projectile reaches a maximum height of 22.5 m.

(c) Determine whether the projectile hits the wall. **[6]**